

# Unit III: Understanding the Interaction

# why interaction matters in UCD?

- It's the bridge between user needs and technology.
- Poor interaction leads to frustration (think of that app that crashes mid-task), while great ones create loyalty (like swiping through Instagram effortlessly).

# Introduction to Interaction Styles & Human Aspects

## Topics

- Device-specific interactions (desktop, mobile, AR/VR, IoT).
- Human aspects: perception, cognition, memory limits, attention.
- Interaction styles overview: Command line, Menu, Form filling, Direct manipulation, Natural language, Conversational UIs.

# Case Study

- Compare MS-DOS CLI vs Windows GUI.
- Siri/Alexa (natural language) vs. ATM Menu navigation.

**Exercise:** List which interaction styles best suit a “Smart Campus App”.

# Examples

- Instagram → Direct Manipulation (swipes, taps).
- Google Search → Command Language (query).
- Food delivery apps → Menu & Form Filling.

# Voice, Visual & Audio-Visual Interfaces

- Voice-based interaction (Alexa, Google Assistant, Chatbots).
- Visual interfaces (dashboards, AR filters).
- Audio-visual interfaces (Zoom, AR/VR meetings, Metaverse).
- Scenarios & Context of Use in interaction design.

# Case Study

- Google Maps voice + visual navigation.
- Duolingo app using visuals + sound for learning.

**Exercise:** Students imagine using an app without visuals (only audio) → e.g., designing for visually impaired users.

# Examples

- YouTube captions (visual + audio).
- VR Gaming (Oculus Quest).
- Zoom auto-transcription.

# Scenarios, User Journey & Mapping

- Scenarios in design (describing tasks, user context).
- User journey & journey maps (steps, pain points, touchpoints).
- Critical task lists for design.

# Case Study

- Uber user journey (booking to drop-off).
- Amazon shopping cart scenario (search → checkout).

**Exercise:** Students create a journey map for “Ordering food via Zomato/Swiggy”.

# Examples

- Student Portal login → accessing grades.
- Netflix journey → “Discover → Select → Watch → Rate”.

# Advanced Interaction Styles & Collaboration

- Interaction devices: Touchscreens, sensors, wearables.
- Collaboration & social media participation.
- Direct Manipulation & Virtual Environments.
- Future of interaction: Metaverse, Brain-Computer Interfaces.

# Case Study

- Google Docs (collaborative interaction).
- Instagram Live → interaction with likes, comments, polls.

# Examples

- VR Meeting in Microsoft Mesh.
- Smartwatch interaction (fitness tracking).
- Slack/Discord → teamwork platforms.

# Mapping the User Journey

- The art and science of user journey mapping.
- It's a visual representation of the user's process of achieving a goal, including their actions, thoughts, and emotions.
- Components of a journey map: phases, actions, feelings, and opportunities.
- **Example:** a user journey map for a user trying to book a movie ticket online, from hearing about the movie to watching it on screen.

# Interaction Styles: Direct Manipulation & Forms

***Direct Manipulation:*** This is where the user directly interacts with the on-screen objects. Think of dragging a file to the trash can.

***Case Study:*** The first Macintosh. Its introduction of drag-and-drop and a graphical interface was a revolutionary step in direct manipulation.

***Menu Selection, Form Filling, and Dialog Boxes:*** While seemingly simple, these have their own design principles. We'll discuss best practices for clarity, efficiency, and error prevention.

- The interface is no longer a single user-to-system dialogue but a multi-user-to-system-to-user dialogue.
- **Social Presence:** Making users aware of others (e.g., read receipts, typing indicators, live avatars - Figma, Google Docs).
- **Concurrency:** What happens when two people edit the same thing at once? (Conflict resolution).
- **Governance:** How are rules enforced? (Upvotes/downvotes on Reddit, reporting systems).
- **Social Translucence:** Designing systems that make social information visible to create accountability and awareness.

# Collaboration & Social Participation

- Designing for a social world.
- How do we design systems that allow users to work together seamlessly? Think of Google Docs or GitHub.
- We'll analyze the interaction styles of platforms like Instagram and TikTok. What makes them so engaging?
- **Case Study:** The "Like" Button. This seemingly simple icon has a complex history and a profound social impact. We'll discuss its evolution and why its design is so effective.

# Case Study: Google Docs

- The cursor of other users is visible (Social Presence).
- Edits happen in near real-time (Concurrency).
- The "Suggesting" mode and version history elegantly solve conflict.
- The chat and comment features are built directly into the interaction flow.

Don Norman suggests organizing screens and menus functionally, designing commands and menu choices to be distinctive, and making it difficult for users to take irreversible actions. Norman also says to provide feedback about the state of the interface (e.g., changing the cursor to show whether a map interface is in zoom-in or select mode) and designing for consistency of actions (e.g., ensuring that Yes/No buttons are always displayed in the same order). State one example you have seen where you know these rules have been violated. Although this is crucial to a user interface's success, suggest why there may be challenges to implement some of Norman's guidelines.